

VAGGELIS GKOUZELOS

Junior AI / Machine Learning Engineer | Mathematics Graduate

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PROFESSIONAL SUMMARY

Mathematics graduate, MSc student in Artificial Intelligence & Software Development at the University of Piraeus (since October 2026) and currently taking the AI Specialist certificate at NKUA. Has built five projects in Python, mostly around solar energy: a machine learning model that forecasts the output of a real solar park, a battery simulation for curtailment losses, an LLM agent with tool use for EV chargers on the grid, a GenAI clinical intake assistant and a solar monitoring tool in beta. Background in linear algebra, statistics, numerical analysis and algorithms. Uses AI tools daily in a data operations role. Looking for a junior role in AI/ML or Python development.

TECHNICAL SKILLS

Programming & Data: Python (pandas, scikit-learn, SciPy, requests), SQL (basic), REST APIs, Git & GitHub, VS Code
Machine Learning (in progress): regression, Random Forest, model evaluation; studying neural networks, NLP, LLM architectures
Generative AI & LLMs: Claude Code, Claude (Anthropic) API, tool use, agent loops, prompt design
Automation & Energy: n8n (workflow automation), pvlib, pandapower, Huawei FusionSolar API
Mathematics: linear algebra, probability & statistics, numerical analysis, discrete mathematics, algorithms
Tools & Languages: Advanced Excel (macros, pivot tables), ECDL · Greek (native), English (B2, fluent written and spoken)

PROJECTS

Source code: github.com/VaggelisGkouzelos (pv-forecast, bess-agent, EVGridCopilot and medintake-ai are public)

pv-forecast – Solar Power Forecasting · github.com/VaggelisGkouzelos/pv-forecast 2026

- ML model forecasting the hourly output of a real 500 kW solar park from weather data (Random Forest, R^2 0.94 on unseen days)
- Python · pandas · scikit-learn · daily forecast automated with n8n and sent via Telegram

bess-agent – Battery Storage Simulation · github.com/VaggelisGkouzelos/bess-agent 2026

- Full-year simulation of a 1,000 kWh battery beside a 500 kW solar park under curtailment; pvlib model validated on real data
- The best charging rule saves 2.4× more energy · Python · pvlib · scikit-learn · SciPy (linear programming)

EVGridCopilot – LLM Agent with Tool Use · github.com/VaggelisGkouzelos/EVGridCopilot 2026

- AI agent that finds how many EV chargers a medium-voltage grid node can host, using a pandapower power-flow tool
- On 10 test questions: LLM alone 0/10 correct, agent with the tool 10/10 · Python · Claude (Anthropic) API · tool use / function calling

medintake-ai – Clinical Intake Assistant · github.com/VaggelisGkouzelos/medintake-ai 2026

- Structures the information a patient provides into a summary with a suggested priority level for the clinician
- n8n · Claude (Anthropic) API · prompt design for structured output

Voltgaia.gr – Solar Monitoring Tool · live beta at voltgaia.gr 2026

- Compares measured output with an independent irradiance model; attributes each loss (grid, O&M, soiling) and prices it in euros
- Python · JavaScript · Huawei FusionSolar API · weather data

WORK EXPERIENCE

Data Operations & Digitisation 09/2025 – Present
MTC – Makedoniki Technical Center · Thessaloniki, Greece

- Use AI tools to automate repetitive documentation work and to run error checks on records
- Maintain Excel trackers to monitor progress and detect inconsistencies across datasets
- Support the digitisation of technical archives into the company's information systems

Back Office Officer 09/2024 – 06/2025
Hellenic Armed Forces · Cyprus

- Maintained high-volume records under strict procedural and confidentiality requirements
- Checked every record before submitting it

Shift Leader 2023 – 2024
Kayak · Santorini, Greece

- Led a shift team in a busy seasonal food & beverage business: staff coordination, scheduling and daily problems

EDUCATION & CERTIFICATIONS

MSc in Advanced Computer Systems – Artificial Intelligence & Software Development 10/2026 – Present
University of Piraeus · Track: Artificial Intelligence and Intelligent Software Applications
Coursework: Advanced Python Programming, Artificial Intelligence, Pattern Recognition & ML, Web Development, Graph Theory

AI Specialist – Certificate of Continuing Education In progress
National and Kapodistrian University of Athens (NKUA) · Machine learning, neural networks, NLP, LLM architectures

BSc in Mathematics 2019 – 2024
University of Ioannina · Algorithms, numerical analysis, statistics, discrete mathematics, linear algebra